Andrés Felipe Berrío Cartagena

Audio Engineer | Perceptual Evaluation of AI Audio | Audio Educator
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PROFILE

Audio engineer working with audio since 2012 and professionally since 2020 (recording, mixing, mastering, sound design), with nearly three years teaching university audio courses and an MSc in Audio and Music Technology from the University of York. I specialise in the **perceptual evaluation of machine-learning audio**: my MSc research used an ABX listening study with experienced listeners to test whether neural captures of a guitar amplifier, an EQ and a compressor are audibly different from the original hardware. Hands-on with audio editing and restoration, spatial/immersive audio, and MATLAB/Python analysis. Native Colombian Spanish; professional English.

EDUCATION

MSc Audio and Music Technology — University of York, UK Sep 2025 – Sep 2026
Awarded with Distinction, October 2026 (award mark 79) · Supervisors: Dr Michael McLoughlin, Prof Helena Daffern

- **Dissertation — Perceptual Evaluation of Neural Audio Device Modeling:** Mark 78/100 (Distinction, UK top band, 70+). ABX listening study with experienced listeners and equivalence testing, comparing Neural Amp Modeler captures of a guitar amplifier, an equaliser and a compressor against their hardware. Objective error did not predict audibility: the amp capture was perceptually equivalent, the EQ capture was borderline and the compressor was clearly distinguishable. Journal article in preparation with the supervisors.
- **Distinction-level marks (UK top band, 70+) in all five graded taught modules:** Immersive and Interactive Audio · Audio Signals and Psychoacoustics · Modelling and Analysing Sound and Music Signals · Sound Interactions in the Metaverse · Project Development.

BA Recording Arts and Music Production — Institución Universitaria ITM, Medellín Feb 2017 – Apr 2022

- **Thesis:** “Así suena Urrao, radiografía sonora” — Honourable Mention for outstanding thesis work.

PROFESSIONAL EXPERIENCE

Sound Engineer — Montaña Producciones, Urrao / Medellín, Colombia Jan 2020 – Present

- Studio and location recording, mixing and mastering for independent and traditional-music projects.
- Audio editing and sound design for audiovisual work; small-format live sound.
- Direction of audiovisual projects with social impact.

Student Welcome Intern (part-time) — University of York, UK Apr 2026 – Sep 2026

- Supported two university-wide working groups on student induction and belonging: meeting minutes, action and progress tracking, and updates to senior staff.

Adjunct Professor — Institución Universitaria ITM, Medellín Feb 2024 – Jun 2025

- Taught Mixing, DAWs II, Perceptive Audio and Critical Listening, and Digital Audio Editing for Video.
- Designed course sessions based on each syllabus; coached students through hands-on recording and editing work.

Adjunct Professor — Tecnológico de Artes Débora Arango, Envigado Aug 2022 – Nov 2023

- Taught Mixing, Recording 1 and 2, MIDI 2, Audio for Video Games, Foley, Musical Preproduction, Corrective Editing, Creative Editing and Audio Computing.

SELECTED PROJECTS

- **Binaural radio drama —** static and moving sources in azimuth and elevation, rendered with SADIE II binaural impulse responses (overlap-add, response smoothing, effects before spatialisation); informal listening check of externalisation and localisation.
- **WANKA —** gestural instrument: webcam hand tracking (MediaPipe, TouchDesigner) plays a digital pututu inside a head-tracked ambisonic-to-binaural model of the Doble Ménsula gallery at Chavín de Huántar (SPAT5, Max/MSP); ODEON room modelling and ISO 3382-1 analysis.
- **SoundSpells —** VR game for PICO 4 (Unity, Steam Audio): gesture-cast audio effects on spatialised sound sources, with session-telemetry analysis.
- **Onset detection with neural source separation —** spectral-flux onset detector on full mixes vs Demucs drum stems (MATLAB; Demucs via Python); F1 0.61 → 1.00 on a dense mix.
- **Virtual-analogue high-gain guitar amp —** MATLAB: TS9-style overdrive, four-stage waveshaper preamp, Bassman tone stack (DAFX code) and cabinet IR convolution.

- **SphereMix —** iPad mixing app built in Swift / AVAudioEngine.

TECHNICAL SKILLS

- **Audio production:** recording, mixing, mastering, dialogue and corrective editing, noise reduction and restoration, Foley, audio for video and games.
- **DAWs and tools:** Pro Tools, Reaper, Studio One, iZotope RX, FabFilter, Max/MSP, SPAT5, ODEON, Unity / Steam Audio.
- **Evaluation and analysis:** listening test design (ABX, equivalence testing; working knowledge of MUSHRA / ITU-R BS.1534), mixed-effects models (GLMM), spectrogram and loudness analysis; statistics in MATLAB, data handling in Python.
- **Audio ML and DSP:** Neural Amp Modeler (NAM) capture and training, Demucs, convolution, filters, ambisonics and binaural rendering, Swift.
- **Languages:** Spanish (native, Colombian) · English (professional; MSc completed in the UK).

AWARDS

- **AES Latin American Student Recording Competition (LASRC), Traditional Music Studio:** Gold (Cali, 2022); Silver (Medellín, 2021; Rio de Janeiro, 2020; Lima, 2019).
- **Santander Career Support Award,** University of York.

PROFESSIONAL DEVELOPMENT

- **Teaching Diploma (120 hours),** Institución Universitaria ITM, 2022.
- **ProCIF undergraduate research group, ITM (2020–2022):** mixing engineer for compilation album #2 (urban genre).